

Analysis of Switching Loss Reduction of SiC MOSFET in Presence of Antiparallel SiC Schottky Diode

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Abstract—SiC MOSFET based high-frequency converter uses device intrinsic body diode during dead-band period for free-wheeling action. This intrinsic body diode has high forward voltage and reverse recovery charge requirement which results in higher switching loss. So SiC Schottky barrier diode (SBD) is connected in anti-parallel configuration across the SiC MOSFET to improve the switching performance owing to its zero reverse recovery characteristics. However switching at high voltage causes an inrush current due to device output capacitance, which increases the overall switching loss. It is essential to differentiate the inrush current due to parasitic capacitance and reverse recovery current of anti-parallel diode, for accurate estimation of reverse recovery loss. This paper investigates the effect of SiC SBD on switching loss in different half-bridge configurations. For analysing the switching losses, clamped inductive load test has been performed at 600 V. Experimental results are provided to validate the switching loss reduction in configurations where SiC SBD is connected in anti-parallel with SiC MOSFET.

Index Terms—SiC MOSFET, SBD, parasitic inductance, parasitic capacitance, reverse recovery, switching loss.

I. INTRODUCTION

Wide band-gap semiconductor devices such as Silicon carbide (SiC) is gaining popularity over its Silicon-based counterparts due to its superior physical and electrical characteristics like high energy band gap (eV), breakdown field (MV/cm) and thermal conductivity (W/cm.°C) [1]. These devices also have lower Drain-Source on-state resistance ($R_{DS,on}$), parasitic capacitance (input, output and reverse transfer capacitance) and reverse recovery charge (Q_{rr}), which enables the converters to work at very high frequency with a reduced switching loss compared to Si devices. Having faster turn-on and turn-off switching rate i.e. high dV_{ds2}/dt and dI_{d2}/dt deteriorates the performance of the device in presence of parasitic capacitance and inductance.

Similar to Si, SiC MOSFET also contains an intrinsic PiN body diode, but the performance of this PiN diode is better, owing to its lower reverse recovery charge requirement. In spite of the improved performance, SiC MOSFET body diode is still the most significant switching loss contributor, and the performance degrades at high temperature high current conditions [2]. Intrinsic body diode of SiC MOSFET induces two types of losses in a half bridge, first is the reverse recovery

loss of body diode, and second is the loss due to addition of peak reverse recovery current on to its complementary MOSFET channel current [3]. Due to this, manufacturers recommend using an external SiC Schottky barrier diode connected in anti-parallel configuration across SiC MOSFET, as it is having zero reverse recovery characteristics.

In this paper the effect of SBD used as an external free-wheeling diode along with SiC MOSFET is investigated. Three different phase leg configuration has been made for comparing the switching losses. In first configuration two SiC MOSFET are connected in series and in second configuration a SiC SBD is connected in anti-parallel configuration with the top SiC MOSFET. In the third configuration, top SiC MOSFET is replaced by SiC SBD. Detailed schematic of various phase leg configurations is shown in Fig. 2. The rest of the paper is organised as follows. *Section II* describes the effect of free-wheeling diode along with device parasitics on switching losses. *Section III* describes the experimental set-up. *Section IV* discusses experimental results of device under test (DUT), obtained from clamped inductive load test and *Section V* concludes the paper.

II. EFFECT OF PARASITICS ON SWITCHING LOSS

Losses in a semiconductor device can be classified into two categories: switching loss and conduction loss. Conduction loss is mainly confined by the load current and the on-state resistance of the SiC MOSFET. Switching loss depends on the load current, voltage blocking and transition time, it also increases due to device parasitics. In a half bridge configuration, the peak power which will be dissipated in SiC MOSFET is dependent upon the maximum current which will flow through its channel, which in turn, is dependent upon the peak reverse recovery current (I_{rr}) of its complementary switch. These switching losses are usually calculated by conducting a clamped inductive load test. In this test, turn-on characteristics are derived from the second turn-on event, and the turn-off characteristics are obtained from the first turn-off event as shown in Fig. 1.

There are three parasitic capacitance present in a MOSFET: gate-source capacitance (C_{gs}), drain-source capacitance (C_{ds})

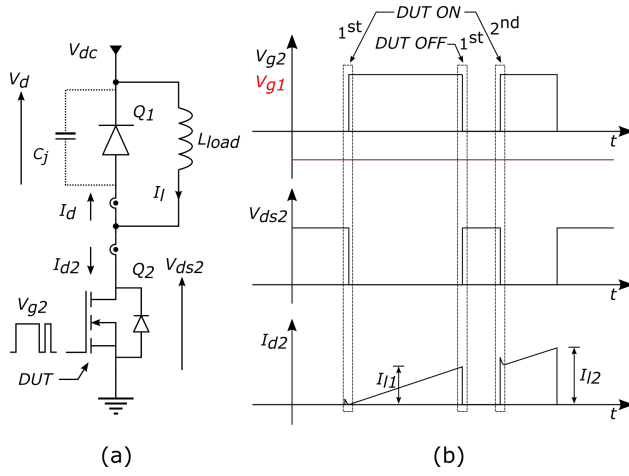


Fig. 1. Switching waveforms in clamped inductive load test circuit

and gate-drain capacitance (C_{gd}). These capacitance can be added together as per Eq. 1 to get the output capacitance (C_{oss}), input capacitance (C_{iss}) and reverse transfer capacitance (C_{rss}). These capacitance are non-linear in nature and are voltage dependent [4], [5].

$$\begin{aligned} C_{iss} &= C_{gs} + C_{gd} \\ C_{oss} &= C_{ds} + C_{gd} \\ C_{rss} &= C_{gd} \end{aligned} \quad (1)$$

Similar to a MOSFET, diode in reverse blocking mode can also be modelled as a capacitance (C_{ak}) as per Eq. 2 [6].

At the initial stage when there is no gate pulse, DC voltage is getting blocked by the switch Q_2 and its parasitic output capacitance ($C_{oss2} = C_{gd2} + C_{ds2}$) is charged to V_{dc} . When the DUT is turned on, the voltage across its terminals (V_{ds2}) starts decreasing to its on state voltage drop level, and at the same time voltage across the top switch (Q_1) starts building towards V_{dc} . Due to this dV/dt , a displacement current (I_{cj}) will flow through the effective parasitic capacitance (C_j) formed across top switch Q_2 as shown in Fig. 1 [7], [8]. A linearised charge equivalent model has been made to find out equivalent capacitance as per literature [9] and can be expressed as Eq. 3.

$$C_x(v_{ds}) = \frac{C_{o,x}}{\sqrt{(1 + \frac{v_{ds}}{a_x})} + b_x} \quad (2)$$

$$C_{x,eq} = \frac{1}{V_{DC}} \int_0^{V_{DC}} C_x(v_{ds}) dv_{ds} \quad (3)$$

where : $x = iss, oss, rss, ak$

The value of $C_{o,x}$ is defined when v_{ds} equals to zero, a_x and b_x can be found out by fitting the capacitance (C_x) versus voltage (v_{ds}) curve provided in datasheet [4], [5]. C_{iss} need not to be fitted as its value remains almost constant with respect to v_{ds} and can be directly taken from the datasheet. The value of these curve fitting parameters can be found out

TABLE I
CURVE FITTING PARAMETERS

Capacitance	$C_{o,x}$	a_x	b_x
C_{oss}	1040 pF	3	1.25
C_{rss}	344.2 pF	0.19	1.25
C_{ak}	388.8 pF	2	1.4

TABLE II
DEVICE CAPACITANCE

Device	Capacitance	Value
SiC MOSFET	$C_{iss,eq}$	950 pF
	$C_{oss,eq}$	123 pF
	$C_{rss,eq}$	12 pF
SiC SBD	$C_{ak,eq}$	35 pF

from Table I. Fig. 3 shows the comparison between the fitted and datasheet provided capacitance versus v_{ds} curve for C_{oss} and C_{ak} .

Equivalent capacitance C_j at Q_2 depends upon the circuit configuration, because SiC SBD and the load inductor has their own parasitic capacitance. But it should be noted that there will be no current due to the parasitic inter-turn capacitance of load inductor (C_l) during the test conditions as load inductor current (I_l) is remaining constant in due process [7]. So for phase leg configuration A [Fig. 2(a)] the effective capacitance will be $C_j = C_{oss1}$, for configuration B [Fig. 2(b)] the effective capacitance will be $C_j = C_{oss1} + C_{ak}$ and for configuration C [Fig. 2(c)] the effective capacitance will be $C_j = C_{ak}$. The approximated junction capacitive current can be written as:

$$I_{cj} = C_j \frac{dV_d}{dt} \quad (4)$$

Initially as the load inductor (L_{load}) current (I_l) is zero, reverse recovery current due to the top diode (can be SiC MOSFET body diode or SiC SBD depending upon phase leg configuration) is also zero, so only I_{cj} will flow in the phase leg at this instant making a small positive transient current I_{cj} . Energy stored while charging C_j can be written as:

$$E_j(V_{DC}) = \int_0^{V_{DC}} v_{ds} \cdot C_j(v_{ds}) dv_{ds} \quad (5)$$

This stored energy E_j will get dissipated through DUT during turn-off, reducing its channel current. So as C_j increases, turn-off energy loss (E_{off}) decreases. During second turn-on C_j will again charge, but the peak current I_{rr}^* at this instant will be more than I_{cj} , as the I_{rr}^* is contributed both by I_{cj} and peak reverse recovery current (I_{rr}) of the top anti-parallel diode. The actual peak reverse recovery current can be found out by Eq. 6.

$$I_{rr} = I_{rr}^* - I_{cj} \quad (6)$$

III. TEST SETUP

For understanding the dynamics characteristics of the SiC MOSFET in hard switching conditions, clamped inductive

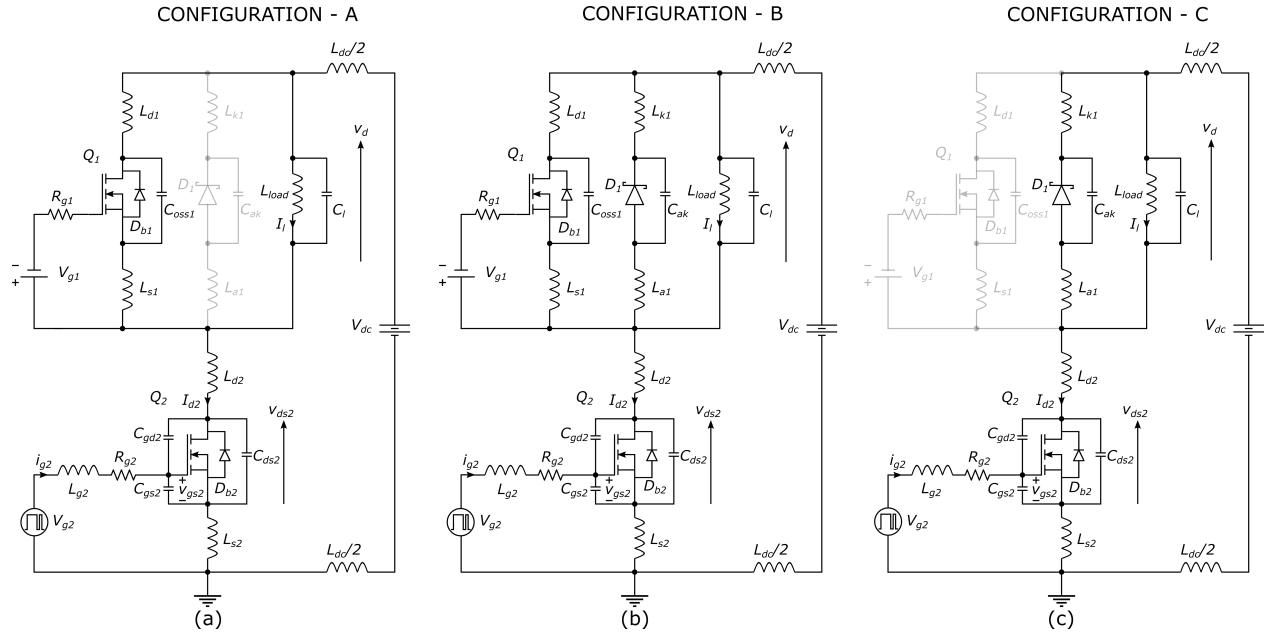


Fig. 2. Schematic of clamped inductive load test including parasitics (a) two SiC MOSFET, (b) two SiC MOSFET and an anti-parallel SiC SBD and (c) a SiC MOSFET and a SiC SBD

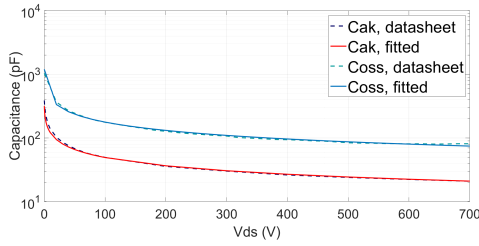


Fig. 3. Fitted curves for C_{oss} and C_{ak} versus V_{ds}

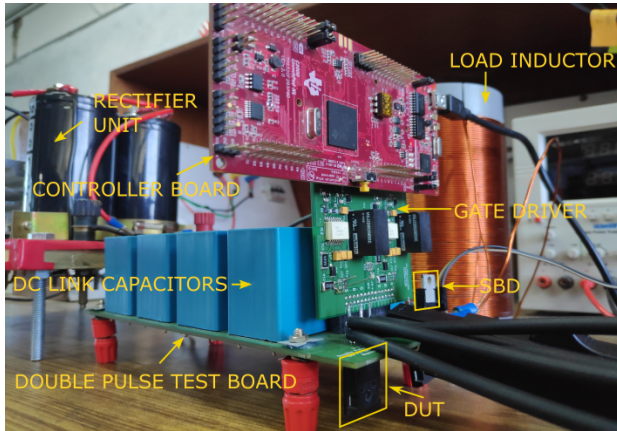


Fig. 4. Clamped inductive load test hardware set-up

load test has been performed. Fig. 2 shows the detailed schematic diagram of the clamped inductive load test set-up, including parasitics for three different phase leg configurations. $Q_1; Q_2$ denotes a second-generation 1200 V, 36 A

Cree C2M0080120D SiC MOSFET and $D_{b1}; D_{b2}$ denotes the intrinsic body diodes of $Q_1; Q_2$ respectively. D_1 denotes a 1200 V, 19 A Cree SiC SBD C4D05120A. L_{d1} , L_{d2} , L_{s1} and L_{s2} are the lead inductances of drain and source terminals for the respective switches. L_{k1} and L_{k2} are the lead inductance of cathode and anode terminals for SBD. L_{load} is an air-cored single layer inductor with inductance of 475 μ H.

Parasitics, e.g. stray inductance in the system causes an over-voltage during turn-off, which can lead to an unintentional turn-on during the switching event, and to avoid this PCB and gate driver circuit has to be appropriately designed. Four parallel strings each comprises of two series-connected low inductance film capacitors are being used to provide the bulk energy required for the clamped inductive load test. SiC MOSFETs were soldered as close as possible to the PCB for reducing the effect of lead inductance.

For clamped inductive load test, gate pulses were given to the bottom switch, and the top switch was held off by a negative voltage. Two 4.99 Ω series resistance is used as an external gate resistance R_g , so that at turn-on R_g equals to 9.98 Ω and for turn-off R_g equals to 4.99 Ω . ACPL-337J used as a gate driver optocoupler along with an isolated DC-DC converter to give +20V and -5V DC supply. This gate driver optocoupler has the capacity to clamp 3 A of miller current, which will mitigate the effect high dv/dt during turn-off and its inbuilt DESAT protection, which was set to trigger at 35 A. The complete test set-up is shown in Fig. 4.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

In order to compare the transient performance for different phase leg configurations, test has been carried out at 600 V DC bus voltage and the current was varied from 0 to 16 A

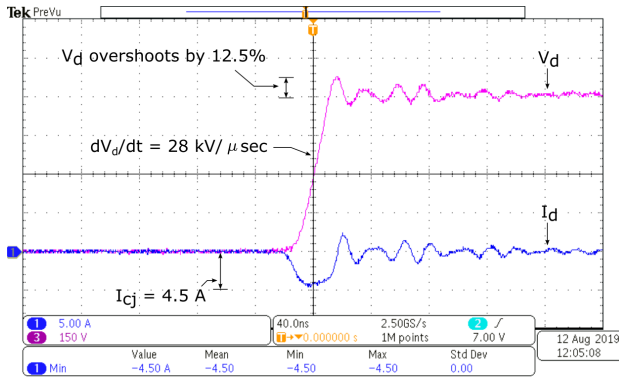


Fig. 5. Phase leg configuration A : Parasitic capacitance current and voltage across diode waveforms during first turn-on. Scale: $v_d \Rightarrow 150V/div$, $i_d \Rightarrow 5A/div$, Time $\Rightarrow 40nsec/div$

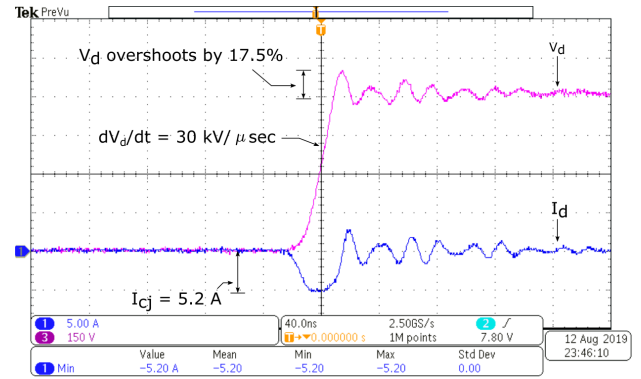


Fig. 7. Phase leg configuration B : Parasitic capacitance current and voltage across diode waveforms during first turn-on. Scale: $v_d \Rightarrow 150V/div$, $i_d \Rightarrow 5A/div$, Time $\Rightarrow 40nsec/div$

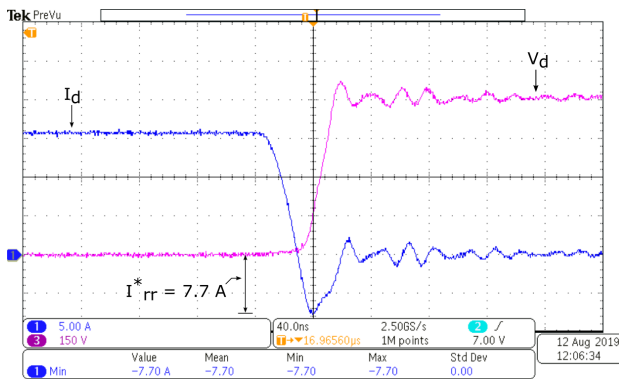


Fig. 6. Phase leg configuration A : Parasitic capacitance current and voltage across diode waveforms during second turn-on. Scale: $v_d \Rightarrow 150V/div$, $i_d \Rightarrow 5A/div$, Time $\Rightarrow 40nsec/div$

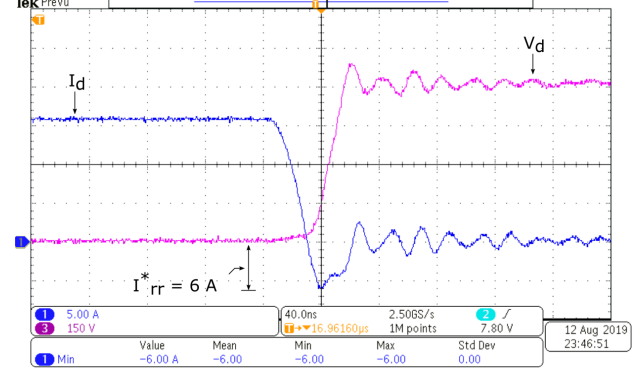


Fig. 8. Phase leg configuration B : Parasitic capacitance current and voltage across diode waveforms during second turn-on. Scale: $v_d \Rightarrow 150V/div$, $i_d \Rightarrow 5A/div$, Time $\Rightarrow 40nsec/div$

at room temperature. Generation of gate pulses were done by a TMS320F28379D micro-controller. Width of these two gate pulses can be estimated using Eq. 7. First gate pulse will determine the current level for which the test is being conducted and the second pulse will make sure that the SiC device is not crossing its safe operating limit. In order to increase the DUT current upto 16 A, a first pulse of width 12 μ sec second was given and a second pulse of width 4 μ sec was given to limit the device current.

$$V_L = L_{load} \frac{dI_{Load}}{dt} \quad (7)$$

TABLE III
CALCULATED AND EXPERIMENTAL RESULTS

Configuration	$C_{j,cal}$	$E_{j,cal}$	$C_{j,exp}$	$E_{j,exp}$
A	120 pF	43.2 μ J	155 pf	33.68 μ J
B	157 pF	57 μ J	174 pf	40.74 μ J
C	35 pF	14 μ J	74 pf	13.35 μ J

A. Intrinsic Body Diode of 1.2 kV SiC MOSFET

Fig.5 shows that at the first turn-on event current through top switch Q_2 is zero but the instant when the DUT starts to conduct, voltage across top switch body diode starts to rise with a rate of 28 kV/ μ sec. This dV_d/dt causes an 4.5 A peak parasitic capacitive current I_{cj} . Here the equivalent C_j was calculated to be 155 pF. At first turn-on event, I_{d2} was at 16 A making the body diode to free-wheel this current during turn-off of DUT. Resistance across the free-wheeling path is assumed to be zero. During second turn-on, body diode of Q_2 goes into reverse recovery region, making I_{rr}^* equals to 7.7 A, but the actual reverse recovery current is 3.2 A as per Eq. 6. So the actual reverse recovery loss happens in the Q_1 anti-parallel diode will be less than it seems to be. Switching energy losses at three different loading conditions are given in Table IV. Experimental results are shown in Fig. 5 and Fig. 6.

B. Intrinsic Body Diode of 1.2 kV SiC MOSFET with 1.2 kV SBD

Similar to previous case, this configuration results a higher peak parasitic capacitive current I_{cj} of 5.2 A, due to the addition of C_{ak} . Here dV_d/dt is found to be 30 kV/ μ sec the

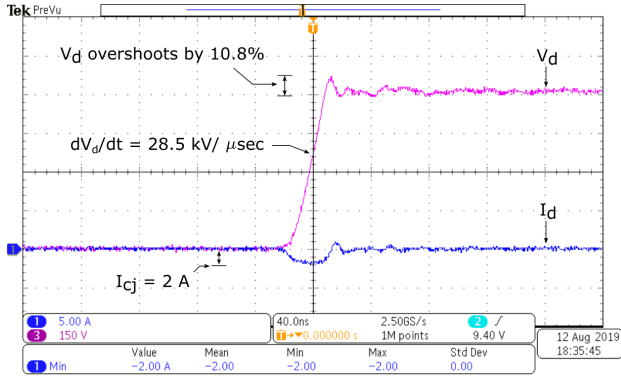


Fig. 9. Phase leg configuration C : Parasitic capacitance current and voltage across diode waveforms during first turn-on. Scale: $v_d \Rightarrow 150V/div$, $i_d \Rightarrow 5A/div$, Time $\Rightarrow 40nsec/div$

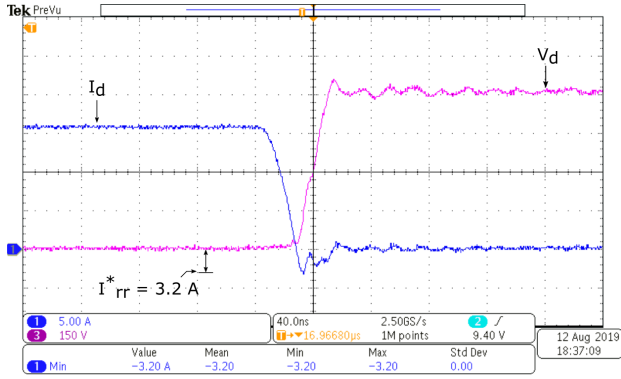


Fig. 10. Phase leg configuration C : Parasitic capacitance current and voltage across diode waveforms during second turn-on. Scale: $v_d \Rightarrow 150V/div$, $i_d \Rightarrow 5A/div$, Time $\Rightarrow 40nsec/div$

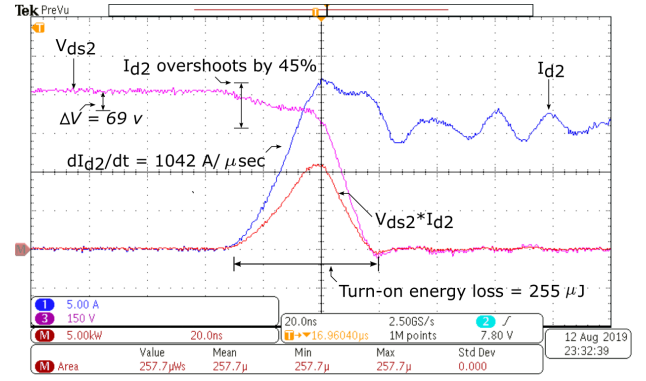


Fig. 11. Turn on voltage, current and power waveforms for Configuration B. Scale: $v_{ds2} \Rightarrow 150V/div$, $i_{d2} \Rightarrow 5A/div$, $P_{sw,off} \Rightarrow 5kW/div$, Time $\Rightarrow 20nsec/div$

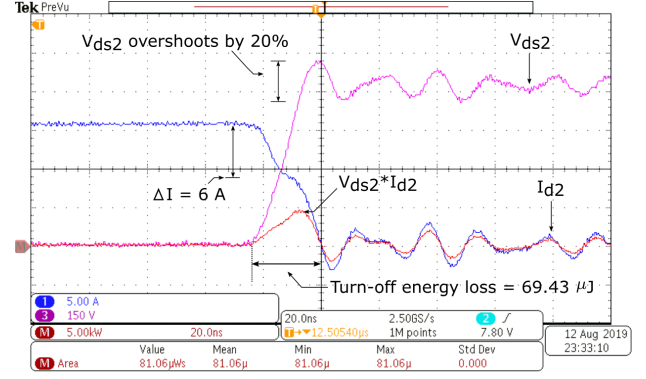


Fig. 12. Turn off voltage, current and power waveforms for Configuration B. Scale: $v_{ds2} \Rightarrow 150V/div$, $i_{d2} \Rightarrow 5A/div$, $P_{sw,off} \Rightarrow 5kW/div$, Time $\Rightarrow 20nsec/div$

equivalent C_j was calculated to be 174 pF. Difference in C_j between configuration A & B comes out to be 19 pF which is approximately similar to the datasheet value of SBD. In the first turn-off event, though there are two SiC diodes in parallel, only SBD will work as a freewheeling diode because of the substantial difference in forward voltage between two diodes, resulting I_{rr}^* to 6 A at second turn-on. Actual peak reverse recovery is only 0.8 A as per Eq. 6, which is negligible. Reduction of I_{rr}^* between first two configuration is 22 % and this reduction also reflects on having a lower E_{on} loss as given in Table IV. Fig. 11 & 12 shows the transient behaviour during turn-on and turn-off respectively. From Fig. 12 it can be seen that during turn off V_{ds2} is increasing to V_{dc} , this is discharging the parasitic capacitance C_j and a transient current because of it, is flowing through the DUT resulting a current drop ΔI of 6 A. So as ΔI increases, turn-off energy loss reduces. Experimental results are shown in Fig. 7 and Fig. 8.

C. 1.2 kV SBD

Use of the configuration C is limited as it does not have bi-directional power flow capability, but this also results to achieve the lowest switching energy loss possible. I_{cj} equals

to 2 A and I_{rr}^* comes out to be 3.2 A as shown in Fig. 9 and Fig. 10. As C_j reduces, the peak voltage overshoot also reduces (Fig. 7, Fig. 5 and Fig. 9) across the the device but the turn-off energy loss increases due to the reduction of ΔI .

TABLE IV
SWITCHING ENERGY LOSS

Current	Configuration	E_{on} (μJ)	E_{off} (μJ)	E_{tot} (μJ)
16 A	A	280.1	71.43	351.53
	B	255	69.84	324.84
	C	208.3	82.81	291.11
11 A	A	191	42.37	233.37
	B	181	40.28	221.28
	C	133.7	55.6	189.3
6 A	A	116	35.26	151.26
	B	108.4	35.76	144.16
	C	75.7	36.7	112.4

V. CONCLUSION

SiC MOSFET based converters operating at high switching frequency increase the switching losses, which leads to higher junction temperature. Operation at such high temperature reduces the reliability of the device. To lower the switching

losses, an anti-parallel SiC SBD is used. This paper investigated the effect of such anti-parallel SiC SBD on switching losses in different phase leg configuration by conducting the clamped inductive load test. It is observed that the inclusion of SiC SBD as a free-wheeling anti-parallel diode along with SiC MOSFET reduces the total switching energy loss by 7.6 %. Switching loss is further reduced to 17.2 % when only SiC SBD is used as a free-wheeling diode. Experimental results also confirms that, apart from reverse recovery current, displacement current (I_{cj}) due to the presence of parasitic capacitance (C_{oss} & C_{ak}) of the diode is a significant contributor in switching loss. However, SiC SBD owing to its zero reverse recovery current improves the overall switching performance.

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